Patrick Bauer

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Experience

Software Engineer II, JPMorgan Chase & Co. — Jul 2022 - Present

- Projects: Global Mass Payments, Brazil Insourcing, Brazil Outsourcing, Chile Modernization, Brazil QR Payments, Chile incoming & outgoing wires, Chile tax payments.
- Application Owner - **Java Springboot** payment router to send Brazil pay-out transactions for sanctions, funds control, and reporting deployed on **Kubernetes**.
- Release Lead - Managed release cuts, regression testing, and **stakeholder communication** for major global code releases.
- Tech Lead - Owned projects end to end from translating requirements from business analysts to actionable **Jira** tickets for the team to work, to deploying completed features and supporting the testing team.
- Engineered custom **Model Context Protocol (MCP)** servers and **AI agents** to automate software development life cycle workflows, integrating **Jira**, **Confluence**, and **Bitbucket** to streamline epic analysis, story creation, documentation review, and PR management.
- Created **Java** applications for collecting and storing client information and reference data for Brazil QR code payments.
- Designed scalable, robust, **Java Spring Boot** apps deployed on **Kubernetes, AWS,** and **On-premise** servers using **Jules CI/CD**
- Saved thousands of hours of manual testing time by implementing new **REST API** endpoints to Cross-Border payments apps to enable **automated regression testing** using **Gherkin** and **Cucumber**.
- Implemented regression test suite from scratch to ensure Chile outgoing wires, incoming wires, and tax payment code changes meet business requirements.
- Achieved unit test code coverage of 100% using **JUnit** framework for all new **Java** code.
- DevOps: Provisioned **Kubernetes** infrastructure for the team's development, testing, and production environments.

Software Engineer, Cummings Aerospace — Aug 2021 - Jun 2022

- Developed and tested software for Raytheon Missiles & Defense systems.
- Implemented microservices architecture using abstract **REST APIs** to handle communication using **C++** and **Python**.
- Designed regression tests using **gherkin** & **BDD** framework and implemented unit tests using **C++** & **Google Test**.
- Implemented real-time secure networking messages between embedded **Linux** systems using modern **C++** and **multithreading**.
- Increased test server efficiency by 87.5% by dynamically allocating hardware using **Python** and **MySQL**.
- Acted as backup **scrum master** for my software development team.
- Used **GitHub** & **Bitbucket** for version control, **Jenkins** for automated building & testing, and **Jira** for agile project management.

Junior Software Developer, *NAVAIR* — Mar 2020 - Nov 2020

- Designed, implemented, and tested new features for pilot training systems tools using **Java**.
- Created specification and requirement documentation
- Used **SVN** for version control and **Jira** for agile project management.

Founder, *Affordable Drone Photography* — Jan 2016 - Mar 2020

- Website Development - Designed and maintained a website using **HTML** and **CSS**.
- Drone Pilot - FAA-certified Unmanned Aerial Systems Remote Pilot.

Web Developer, *Bauer International Group* — Jan 2012 - Mar 2020, Jun 2023

- Re-created website in 2023 using **React**, and deployed on **Google Cloud** using automated **Github Actions** pipeline.
- Original website created on WordPress with **HTML** & **CSS**.

Education

BS Computer Science, GPA: 3.94, University of Central Florida **May 6, 2021**
AS Computer Programming & Analysis, Honors, Florida Southwestern State College **May 3, 2019**

Certifications & Licenses

AWS Certified Cloud Practitioner, Amazon **Jan 25, 2023 - Jan 25, 2026**
Secret Security Clearance, DoD **Mar 2020 - Jun 2024**
Unmanned Aerial Systems Remote Pilot, FAA **Apr 17, 2018 - Apr 16, 2020**

Projects

Autonomous Search With AI, *Lockheed Martin*

- Real-time Object Detection Simulation using **ROS** (Gazebo)
- **Machine Learning (AI)** & Autonomous Navigation (**Python**)

Livestream & Automated Treat Dispenser, *Bird Gardens of Naples*

- Set up 24/7 live stream networking for local NPO
- Built and programmed automated bird treat dispenser, hooks into streaming web **API** to dispense bird treats using a **servo** motor when a viewer donates money to the non profit stream: https://www.twitch.tv/birdgardensofnaples
- **Linux** Raspberry pi running 2 **docker** containers: **python** backend, and **nodejs** front end.

Home Linux & Docker Server

- Leveraged my **architecture** & **system administration** skills to deploy a **Linux** home server (private cloud) running 18 **docker** containers including nextcloud for personal photo and video backup, traefik for reverse proxy, gluten for vpn, adguard-home for internet security, and cloudflare-ddns for domain DNS record updates.

Stock Trend Analyzer

- Built data collection & visualization software using **python**, and stock data from public **APIs**
- Backtested and compared algorithms on 10 years of S&P 500 data to compare investment strategies.

Concurrent Blockchain, *University of Central Florida*

- Developed multi-threaded **blockchain** software comparing computing performance of **C++**, **Java**, and **Rust**

Languages & Apps

Java, C++, Python, SQL, GCP, AWS, Kubernetes, Docker, Jules, Github, Bitbucket, Cucumber
English (fluent), Polish (fluent), Spanish (basic)

Clubs & Events

UCF Senior Design Showcase Judge **2021 - 2022**
UF Swamp Hacks - Hackathon **2020**
UCF Programming Club **2019 - 2020**
UCF Coding Competition **2019**